

EXPERIMENT NO. 2

Common Values Between Two Arrays

Programming Language: C

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Aim

To write a C program that counts how many elements of each array also exist in the other array.

Question

You are given two integer arrays `nums1` and `nums2` of sizes `n` and `m`, respectively. Calculate the following values:

`answer1`: the number of indices `i` such that `nums1[i]` exists in `nums2`.

`answer2`: the number of indices `i` such that `nums2[i]` exists in `nums1`.

Return `[answer1, answer2]`.

Example 1

Input: `nums1 = [2,3,2]`, `nums2 = [1,2]`

Output: `[2,1]`

Example 2

Input: `nums1 = [4,3,2,3,1]`, `nums2 = [2,2,5,2,3,6]`

Output: `[3,4]`

Explanation: The elements at indices 1, 2, and 3 in `nums1` exist in `nums2` as well. So `answer1` is 3. The elements at indices 0, 1, 3, and 4 in `nums2` exist in `nums1`. So `answer2` is 4.

Algorithm

Step 1: Start.

Step 2: Store the arrays `nums1` and `nums2`, compute their sizes `n` and `m`, and set `answer1 = 0` and `answer2 = 0`.

Step 3: For each element `nums1[i]`, search `nums2` linearly. If a match is found, increment `answer1` and stop searching for that element.

Step 4: For each element `nums2[i]`, search `nums1` linearly. If a match is found, increment `answer2` and stop searching for that element.

Step 5: Print `[answer1, answer2]`.

Step 6: Stop.

Program (C)

```
#include <stdio.h>

int main() {
    // Example 2 Arrays (Modify these to test other cases)
    int nums1[] = {4, 3, 2, 3, 1};
    int nums2[] = {2, 2, 5, 2, 3, 6};

    int n = sizeof(nums1) / sizeof(nums1[0]); // Size of nums1
    int m = sizeof(nums2) / sizeof(nums2[0]); // Size of nums2

    int answer1 = 0;
    int answer2 = 0;

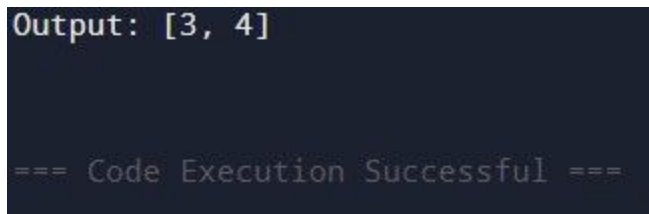
    // 1. Calculate answer1: Count how many elements of nums1 exist in nums2
    for (int i = 0; i < n; i++) {
        // Search for nums1[i] inside nums2
        for (int j = 0; j < m; j++) {
            if (nums1[i] == nums2[j]) {
                answer1++;
                break; // Found it! No need to keep looking for this specific element.
            }
        }
    }

    // 2. Calculate answer2: Count how many elements of nums2 exist in nums1
    for (int i = 0; i < m; i++) {
        // Search for nums2[i] inside nums1
        for (int j = 0; j < n; j++) {
            if (nums2[i] == nums1[j]) {
                answer2++;
                break; // Found it! No need to keep looking for this specific element.
            }
        }
    }

    // Output the result
    printf("Output: [%d, %d]\n", answer1, answer2);

    return 0;
}
```

Output



=== Code Execution Successful ===

Complexity Analysis

Time Complexity: $O(n * m)$ in the worst case, $O(n + m)$ in the best case.

Space Complexity: $O(1)$

Result

Thus the program was executed successfully and the output was verified.